When Constants Win: Lessons from Re-analysing a Privacy-Harm Heuristics Evaluation

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Abstract

We re-analyse a small study that asked whether interpretable heuristics can estimate privacy harm, and how they compare with large language models (LLMs). Its recorded result was that a "rules static" arm beat a plain LLM, a rules-dynamic arm, retrieval augmentation, and two hybrids on all four reported metrics. We made no model calls and used only the committed artifacts plus the recovered per-case outputs of all 18 recorded trials. We find:

- A label-free constant that predicts all four of Solove's harm groups for every case beats the headline arm on Jaccard and nDCG@5. The intervals exclude zero after correcting for 15 comparisons. Predicting no harm ties the arm on exact match.
- On micro-F1, neither the headline arm nor a 2026 rerun is distinguishable from the best constant, even after out-of-taxonomy labels are removed. Against the label-free all-four constant both also tie; only the post-hoc filtered rerun is better.
- Five of the six arms sent an identical prompt. Rules-static, the one arm with a different prompt, did differ from the other five (one-sided $p = 0.007$, unadjusted), but not enough to beat a constant.
- The gold set was LLM-generated and mostly off-topic.
- Every multi-arm result in the original public artifact came from a dry run on a different gold set.
- Every label in the released corpus came from a keyword fallback.

We distil eight evaluation lessons and one data-ethics lesson.

1 Introduction

Estimating the *harm* of a privacy incident, not only detecting that one occurred, matters to regulators, auditors, and engineers deciding what to fix first. Solove's taxonomy organises privacy problems into information collection, processing, dissemination, and invasion [22], and Citron and Solove catalogue the injuries that follow [4]. Recent work applies the taxonomy to AI incidents [11]. A natural question is whether cheap, interpretable heuristics can apply that structure at scale, and whether an LLM does better.

A 2025 master's practicum asked exactly that: *can we estimate privacy harm using heuristics, given facts parsed by a small language model*, and *how do non-language models compare with LLMs and with an LLM applying classical privacy frameworks*? Its public artifact records a headline table in which a "rules static" arm led every metric. We re-examined the evidence offline and asked what it supports. The contributions are:

- a re-analysis against label-free constant predictors. It uses the per-case outputs of every recorded trial and the study's own metric code, with paired, multiplicity-adjusted bootstrap intervals (§3);
- an audit of what the evaluation actually compared: its prompts, scorer revisions, gold set, dry runs, and label sources (§4);
- nine lessons, each tied to its evidence (§5).

2 The original study

Pipeline. Public incident reports (regulator and securities filings, news, forums, Wikipedia) were crawled, labelled with Solove categories, and featurised. Five interpretable model families were trained: a decision tree; an Explainable Boosting Machine [14, 15, 16]; Bayesian rule lists [12]; a sparse linear model; and a Chow–Liu Bayes net [3]. The labeller was to use GLiNER2 [24], a successor to GLiNER [23].

Evaluation. Six arms were scored on a 50-case gold set (the *v3* set): a plain LLM baseline; *rules static*, a fixed rule block in the prompt; *rules dynamic*; RAG; and two hybrids. All were served by Gemini. The metrics were instance-averaged Jaccard, exact-match ratio (EMR), micro-F1, and nDCG@5 [25, 9] over Solove's four groups.

The recorded summary is dated 2025-11-06 and says it averages five trials. Its numbers instead reproduce exactly from three trials per arm run on 2025-11-22, recorded in a trial summary and 18 per-case result files. The public artifact omitted these; they survived in the private predecessor repository, and we republish them with this paper. The trial configurations record no model name. Earlier runs on the same day scored unmapped gold subtypes against group-level predictions, got near-zero overlap, and were superseded by one complete pass. We republish those superseded runs too. All 22 scored against unmapped gold subtypes: ten 50-case runs reached micro-F1 of at most 0.022, and twelve 20-case runs are the prompt-echo failures described in §4. We see no sign of selective reporting.

A 2026 rerun repeated rules-static once. Its model is likewise unrecorded in the results file and known only from README prose. The repository also added an *offline* arm that calls no

LLM and uses the corpus keyword labeller; we recompute it per case.

3 Constants against the arms

An empty-majority gold set. Of the 50 gold cases, 35 carry no harm label. The other 15 carry 35 group labels (processing 13, invasion 11, collection 7, dissemination 4). The scorer counts an empty prediction on an empty case as an exact match but as a Jaccard and nDCG of 0. EMR therefore mostly measures how often an arm predicts nothing, and Jaccard and nDCG are read against a ceiling of 0.30 (Table 1).

Label-free constants. We compare two constants that need no labels to construct: the empty set, and all four groups in the taxonomy's own order. We fixed them while revising the analysis, after the scores of all 65 constants had been computed, so they are label-free but not preregistered. All intervals come from a paired case-resampling bootstrap (4,000 resamples) that averages the three trials inside each resample. We report 15 paired comparisons; Bonferroni-adjusted intervals ($\alpha = 0.05/15$) are in the artifact, and we note below which gaps survive that adjustment.

Against the recorded headline arm. Using its recovered 2025 trials:

- *All four* beats rules-static on Jaccard, by $[0.049, 0.172]$, and on nDCG@5, by $[0.070, 0.231]$. Both gaps survive the Bonferroni adjustment.
- *Empty* ties it on EMR ($[-0.013, 0.093]$) and on Jaccard$_{\emptyset=1}$ ($[-0.091, 0.055]$). On that convention the arm's point estimate (0.721) is higher than the constant's (0.700).
- On micro-F1, the arm minus the best constant, re-selected inside each resample from all 16 label sets, is $[-0.173, 0.214]$: a tie. Against the label-free all-four constant it is $[-0.102, 0.261]$, also a tie.

So the constants beat the headline arm on two metrics and tie it on the other two. The arm is not significantly better than a constant on any of the four set metrics.

Against the 2026 rerun and the offline arm. Against the rerun:

- Empty wins EMR by $[0.22, 0.48]$, and all four wins Jaccard by $[0.033, 0.111]$. Both survive the adjustment.
- All four also wins nDCG@5 by $[0.017, 0.112]$, but that gap's adjusted lower bound touches zero (-0.0001), within Monte Carlo error.
- On micro-F1 the rerun ties the best constant ($[-0.128, 0.071]$) and the all-four constant ($[-0.035, 0.112]$).

The rerun's output contains eight labels outside the gold label space (root-cause terms such as `lack_of_consent`), and each is a guaranteed false positive. Dropping them post hoc lifts its micro-F1 to 0.463, but it still ties the best constant ($[-0.031, 0.217]$). Against the label-free all-four constant, though, it is better: $[0.060, 0.261]$, and $[0.006, 0.314]$ at the adjusted level. That is the only comparison here in which an arm beats a label-free constant, and it is post hoc.

Against the offline arm, all four wins Jaccard ($[0.042, 0.151]$, which survives adjustment), micro-F1 ($[0.007, 0.149]$), and nDCG@5 ($[0.009, 0.146]$).

The labelled cases alone. On the 15 non-empty cases, the all-four constant scores Jaccard 0.583 and nDCG@5 0.790. That is far above the 2025 rules-static arm (0.226 and 0.299) and the 2026 rerun (0.351 and 0.587, or 0.528 and 0.675 filtered). With so few informative cases, the micro-F1 intervals against the best constant are 0.20–0.39 wide, so modest real effects would go undetected [1].

Severity. Severity (1–5) is where the LLM arms show consistent signal. Always predicting the modal score 1 is more *accurate* (0.78) than 2025 rules-static (0.77) or the 2026 rerun (0.50), but it has a quadratic-weighted κ of 0. The arms reach κ 0.40–0.54 in 2025 and 0.33 ($[0.05, 0.56]$) in 2026. The gold severities are themselves inconsistent: four harm-labelled cases, such as "Data warehouse", have severity 1.

4 What the evaluation compared

Five arms, one prompt. In the committed code, the prompt differs by arm only for rules-static. The deterministic-first hybrid adds text only when a deterministic model's output exists, and the runner never supplies one. The prompt builder never reads its context argument, so no retrieved text could reach the RAG arm's prompt. Building each arm's prompt offline yields two distinct texts: one for rules-static, and one shared by the other five. The 2025 trials sit beside an older runner (predecessor commit `d1d38594`). Its prompt builder has the same mode branches, and its runner likewise never passes a deterministic model's output or any context, so the same holds for 2025.

The five identical-prompt arms therefore measure replicate noise, and rules-static does stand out from them. Its paired micro-F1 gain over their mean is $[0.021, 0.221]$ and its EMR gain is $[0.016, 0.133]$. Neither survives the Bonferroni adjustment. A one-sided trial-level permutation test on micro-F1 gives $p = 6/816 \approx 0.007$ (two-sided, 0.020). The arms also ran in consecutive time blocks, so arm is confounded with time of day. Rules-static's outputs differ from those of the other five arms, and the rule block is the only prompt difference. It still did not beat a constant.

The "rules" are five generic compliance sentences, such as "If a data breach occurs, affected users must be notified within 72 hours." They are unrelated to the trained models. Yet the original summary calls the arm "deterministic regex/keyword heuristics", although an LLM produced every one of its labels.

Scorer and prompt changed between runs. The 2025 scorer mapped gold labels to the four groups but scored predictions raw. The current scorer maps both. The 2025 rules-static outputs are already group-level, so they score identically under

	All 50 cases					15 labelled cases	
Arm	Jaccard	Jaccard$_{\emptyset=1}$	EMR	micro-F1	nDCG@5	Jaccard	nDCG@5
Recorded 2025 arms (mean of 3 trials; per-case outputs recovered)							
Rules static	0.068	**0.721**	0.667	**0.391**	0.090	0.226	0.299
LLM baseline	0.056	0.662	0.620	0.285	0.072	0.185	0.239
Rules dynamic	0.055	0.641	0.593	0.281	0.073	0.182	0.244
Hybrid, deterministic first	0.053	0.640	0.593	0.275	0.068	0.178	0.227
RAG	0.047	0.627	0.580	0.242	0.061	0.156	0.203
Hybrid, LLM first	0.036	0.629	0.600	0.195	0.049	0.120	0.164
Later runs							
Rules static, 2026, current scorer	0.105	0.465	0.360	0.336	0.176	0.351	0.587
same outputs, 2025 scorer	0.041	0.401	0.360	0.167	0.041	0.136	0.136
same outputs, filtered to groups (post hoc)	0.158	0.518	0.400	0.463	0.202	0.528	0.675
Offline keyword arm (no LLM)	0.081	0.161	0.080	0.222	0.162	0.269	0.541
Label-free constants							
Always $\emptyset$	0.000	0.700	**0.700**	0.000	0.000	0.000	0.000
Always all four, taxonomy order	**0.175**	0.175	0.040	0.298	**0.237**	**0.583**	**0.790**
Constants, best per column (selected on test labels)							
Best of 65 constants	0.192	0.700	0.700	0.356	0.282	–	–
Perfect predictor (ceiling)	0.300	1.000	1.000	1.000	0.300	1.000	1.000

Table 1: All rows use the same 50-case gold set and the repository's metric code. That code reproduces every recorded 2025 trial and the 2026 run exactly. The 2025 rows are scored as recorded, with predictions unmapped; mapping them would move only the LLM-first hybrid, by about 0.001. It gives empty-vs-empty a Jaccard and nDCG of 0 but an EMR of 1, so a perfect predictor scores only 0.300 on Jaccard and nDCG; Jaccard$_{\emptyset=1}$ uses the other common convention. The 65 constants are every ordered subset of the four groups. Bold marks the best value per column among the recorded arms, the 2026 current-scorer run, the offline arm, and the label-free constants. It excludes the post-hoc filtered row, the test-selected constants, and the ceiling.

both. They contain 59 group labels, 8 others, and no subtypes. Seven of the eight others are group names with text appended, such as "invasion (mass surveillance)", which a normalising parser would have credited (a small instance of L3). The 2026 prompt, revised on 2025-11-29, added a root-cause allowlist and told the model it may put root causes in the same field. Its outputs therefore depend on mapping: Jaccard is 0.041 unmapped and 0.105 mapped.

Under one scorer, the 2025→2026 change raised Jaccard (0.068 to 0.105) and nDCG@5 (0.090 to 0.176) and lowered EMR (0.667 to 0.360) and micro-F1 (0.391 to 0.336). Those differences confound the prompt revision, one trial against three, and possibly the model, which neither run records; the provider default was the same in both. They are not evidence of model drift. Hosted-model drift is real [2, 20], but this artifact cannot measure it.

An LLM-generated, mostly off-topic gold set. The summary records that the v3 gold set was generated with Gemini 2.5 Pro. Arms from the same family may share its errors. That is analogous to the self-preference of LLM judges [18, 26], and it would favour the arms, which makes the constants' results more telling. Of the 50 cases, 47 are general Wikipedia articles; by title, none reports a specific privacy incident. The labelled cases mix on-topic articles (National Security Agency, Browser security, Profiling) with off-topic ones (the Railway Regulation Act 1844, Data warehouse). The JSON lists every title.

No case has the `harms` field the scorer reads first, so it silently falls back to `root_causes`. A code comment says that field holds technical causes, but in this set it holds only Solove subtypes, so the fallback read the right labels under the wrong name. The real contract failure is on the prediction side: the 2026 prompt invites root causes into the field the scorer treats as harms.

A second, 96-case set has labels drafted by `gemini-2.5-flash` (92 of 96) and is marked as reviewed in July 2026. Its review timestamps are bulk writes (median gap 1.4 ms), so they cannot show per-case review. It has no empty-gold cases. By L6 below, it could serve as a benchmark only after independent relabelling.

Dry runs shipped as experiments. The artifact ships fourteen "comprehensive" run directories, the largest spanning 320 configurations, and it omits the live trials behind the headline table. All fourteen name `golden_cases.jsonl` (the file that later became the 96-case set), not the v3 set. Seven are flagged `dry_run`. The five that report per-arm metrics are all dry runs, and within four of them every arm has an identical mean micro-F1; the fifth has two values. The placeholder score, 0.417, is higher than the headline arm's 0.391, on a different gold set. The seven live runs committed no per-arm metrics.

Pairwise-test files show Bonferroni thresholds and Cliff's δ in three directories. One file is truncated and does not parse,

and 15 of the 43 parseable entries are placeholders ($p = 1$, zero difference). A related hazard actually fired. The LLM client turns most API errors into a truncated echo of the prompt, which parses as an empty prediction. On the morning of the recorded trials, twelve 20-case smoke runs (two per arm) failed this way on every case. Each scored EMR 0.65 by predicting nothing, higher than five of the six recorded arms. None of the 18 recorded trials hit it, and the rerun's single parse failure was a malformed model reply.

Heuristic labels in, heuristic rules out. The labeller enables GLiNER2 only when no LLM provider is configured, and it silently swallows a GLiNER2 load failure, falling through to keyword scoring. In the released 33,472-row corpus, every label's recorded source is that keyword fallback. 61.4% of labels are `unknown`, and 6 of the 21 other classes (e.g. `regulatory_violation`) are not Solove categories. The offline arm uses the same labeller and emits 46 predictions from five of those classes. The models were trained on seven features: a log count of affected individuals and six keyword flags. So they approximated one keyword heuristic from another, which cannot answer the study's first question, and they did so poorly:

- Macro-F1 was 0.14 (decision tree), 0.10 (EBM), 0.20 (sparse linear), and 0.04 (Bayes net).
- The three 2026 models share one 984-row test split: 578, 573 and 516 rows correct. The Bayes net stores 984 predictions, all of one class, which is 52.4% of that split but only 2.1% of the release.
- The decision tree's exported rules imply 3,934 training rows, so about 4,918 labelled rows in all, against 33,472 in the release. The release is not the training set, and the training corpus's recorded hash matches no archived copy we could find.
- One of 16 per-category rule-list classifiers trained.
- The exported "heuristics" are corrupted by an exporter bug. Each of the 24 rules reports support exactly $1/3{,}934$ and a leaf size of 0 or 1, because the tree's per-leaf value array held class fractions, not counts. The reported "precision" is the share of class index 1, not of the predicted class. The tree file shows that the first rule ("no keyword present") actually covered 517 of 3,934 training rows (13.1%).

5 Lessons

L1 Fix label-free constant baselines and score them with the same scorer. Use empty, all labels in a fixed order, and the modal score for ordinal targets. Report oracle constants separately, re-selected inside the bootstrap. A metric on which no arm beats the constants supports no claim.

L2 Disclose empty-set conventions and the ceiling they imply. Report the gold empty share, and score the non-empty subset separately.

L3 Field names lie; pin the prompt–scorer contract. Validate predictions and gold labels against one schema, and fail on missing fields or out-of-schema labels instead of falling back.

L4 Diff what each arm actually sends. Arms that differ only in name measure replicate noise. That is useful, but only if you know it.

L5 Make placeholders unable to pass for results. Refuse to aggregate or test arm-invariant outputs, keep dry runs out of results directories, and fail on client errors instead of echoing the prompt.

L6 Keep the gold set independent and its review auditable. Do not generate it with the model family under test, and record who reviewed each case and when.

L7 Measure the label pipeline you ran, not the one you designed. Log the share of labels from each source, and never swallow a model-load failure [21, 17].

L8 Ship the per-case outputs behind every reported number. Record the model version, scorer revision, prompt hash, and trial count in each result file. Here the extraction kept the dry runs and dropped the live trials. Make the analysis deterministic as well: an earlier draft of this one drew a bootstrap candidate set in Python's hash order and changed with the interpreter's seed.

A data-ethics lesson. A pre-release audit found that the corpus's social-media portions republished verbatim, re-identifiable posts in which private individuals described legal, medical, and financial trouble. That is the secondary-use harm the project studied, and a tested scrub removed it before release. Scrub at collection, not at release.

These lessons echo known failures: hypothesis-only and trivial baselines [8, 19], shortcut learning [7], leakage [10], fragile benchmark rankings [5], evaluation failures across ML [13], and significance testing [6]. What is new with LLMs is how cheaply one model family can produce each ingredient (gold labels, rules, predictions, and even the fallback text on error), and so end up on both sides of a comparison.

6 Limitations

- We made no model calls, so we cannot say how the arms would score on the 96-case set.
- The label-free constants were fixed after we had seen the scores of all 65 constants; they are not preregistered.
- The filtered sensitivity check is post hoc.
- With 15 informative cases, statistical power is low.
- Our judgement that the gold articles are off-topic rests on titles.
- The 2025 trial files come from a private repository. We republish them together with the 22 superseded runs, and the 2025 scorer reproduces all 18 recorded trials exactly.
- The gold file is byte-identical to its Nov-2025 version, and the taxonomy that maps its subtypes to groups is unchanged.

7 Conclusion

Label-free constants beat the recorded headline arm on Jaccard and nDCG@5 and tie it on exact match and micro-F1. No arm, including a 2026 rerun with its out-of-taxonomy labels removed, is significantly better than the best constant on any

of the four set metrics; only that post-hoc filtered rerun beats a label-free constant, on micro-F1. Rules-static's outputs do differ from the identical-prompt arms, and the arms carry real signal on severity. But the study's first question, whether heuristics can estimate harm, remains open, because every label in the released corpus was itself a keyword heuristic, and the training set's own labels cannot be checked.

The study's uncertainty machinery (confidence intervals, corrected tests) ran only on dry runs against a different gold set, and its headline table carried no uncertainty at all. What it offers is a compact, reproducible case of evaluation failure, and nine concrete ways to avoid repeating it.

Artifact availability

The repository, including gold sets, recorded runs, the recovered 2025 trials and superseded runs (`paper/trials_2025/`), trained models, and a scrubbed corpus (release `v1.0.0`), is public at `https://github.com/Savage-Fred/privacy-harm-heuristics`. `paper/reanalysis.py` regenerates `paper/reanalysis.json`, the source of every re-analysis number, offline. It asserts that it reproduces the 18 recorded 2025 trials, the recorded table, the 2026 rerun, and the offline arm exactly. Its output is identical under different `PYTHONHASHSEED` values.

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